

University Student Management System

PROJECT PROPOSAL

1. INTENDED USE OF THE SYSTEM

The University Student Management System is a centralized web-based system designed to support and organize the main academic operations of a university. The system is intended to provide a single platform where students, instructors, college administrators, and university/system administrators can access and manage the information and services related to their roles.

Students:

Students will use the system as a self-service platform to manage their academic activities. They will be able to view their personal and academic information, access available courses, select courses during registration, and complete their course enrollment. The system will also allow students to view their schedules, attendance records, grades, semester GPA, cumulative GPA, and academic transcript. Students will be able to generate and download their registration forms and transcripts when needed. These functions provide students with easier access to their academic information without depending on separate paper-based records or manual processes.

Instructors:

Instructors will use the system to manage their academic responsibilities and interact with the students enrolled in their courses. They will be able to view their assigned courses and schedules, access enrolled student lists, and manage academic advising. The system will provide instructors with tools to record and manage student attendance for individual lectures and to enter and manage grades for their courses. Instructors will also be able to view the required academic information needed to perform their teaching and advising responsibilities.

College Administrators:

College Administrators will use the system to manage academic information and operations at the college level. Their responsibilities include managing departments and courses, defining academic rules and credit-hour requirements, managing examinations, and managing student results. They will also be able to view and manage college-level statistics, including information related to departments, academic levels, and student success rates.

University / System Administrators:

University/System Administrators will use the system to manage information at the university level. They will be responsible for managing colleges and college administrators, viewing college information, and monitoring university-wide statistics. The system will also provide university administrators with an overview of students, departments, courses, and instructors within each college.

Overall, the system is intended to provide each user with role-based access to the functions and information relevant to their responsibilities. Authentication, authorization, and role-based permissions will help ensure that users can access the appropriate parts of the system according to their assigned roles.

2. OVERALL FUNCTIONALITY

The University Student Management System will integrate several academic and administrative functions into one system. Its overall functionality is designed to organize academic data, automate common academic processes, and provide users with easier access to the information required to complete their tasks.

Authentication and User Management Module

The system will provide an authentication and user management module that allows users to log in securely and access the system according to their roles. It will support role-based authorization and redirection, together with basic permission control. The system will also provide a common application layout containing the main navigation elements and user profile functionality.

Student Registration Module

The Student Registration module will allow students to view available courses, select the courses they want to register for, and view their selected courses and total registered credit hours. The system will perform course enrollment and schedule conflict detection before completing the registration process. After registration, the system will provide a registration confirmation and generate a registration form containing the student's information, registered courses, and credit hours.

Instructor and Attendance Management Module

The Instructor and Attendance Management module will allow instructors to manage their courses and lectures and record student attendance. For each lecture, instructors will be able to view enrolled students and mark them as Present or Absent. Attendance records will be saved, and the system will allow students to view their attendance percentage and attendance details. Completed lecture attendance will become read-only to maintain the integrity of attendance records.

Grades, GPA, and Transcript Module

The Grades, GPA, and Transcript module will support the entry and management of student grades. Instructors will be able to enter grades and grade components, while the system will validate the entered grades and calculate total grades. The system will calculate semester GPA and cumulative GPA and organize student grades by academic term. Students will be able to view their grades, GPA, academic history, and transcript, with options to print or download their transcript.

Administration and Academic Management Module

The Administration and Academic Management module will provide university-level and college-level administrative functions. University administrators will manage colleges and college administrators, while college administrators will manage departments, courses, academic rules, examinations, and results. The system will support examination scheduling, exam-room assignment, instructor or committee assignment, result correction, result approval, and result publication. It will also provide statistics related to departments, academic levels, passed and failed students, and success rates.

By bringing these functions together in one centralized system, the University Student Management System will help its users complete academic and administrative tasks in a more organized and accessible way. The system will reduce the need to manage academic information through separate processes and provide students, instructors, and administrators with access to the information and functions required for their respective responsibilities. The final system will integrate all modules and provide a consistent workflow across student, instructor, college administration, and university administration activities.